

Problem

A 100-mH inductor is connected in parallel with a 2-k Ω resistor. The current through the inductor is $i(t) = 35e^{-400t}$ mA. (a) Find the voltage v_L across the inductor. (b) Find the voltage v_R across the resistor. (c) Does $v_R(t) + v_L(t) = 0$? (d) Calculate the energy stored in the inductor at $t = 0$.

Step-by-step solution

Step 1 of 5

Consider the following circuit of a 100 mH inductor connected in parallel with a 2 k Ω resistor.

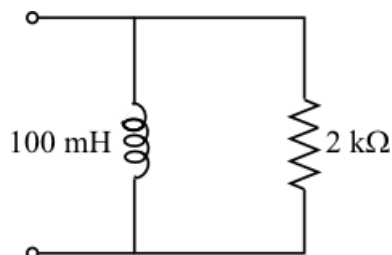


Figure 1

Step 2 of 5

The current through the inductor is,

$$i(t) = 35e^{-400t} \text{ mA}$$

(a)

Determine the voltage across the inductor.

$$\begin{aligned} v_L(t) &= L \frac{di(t)}{dt} \\ &= (100 \times 10^{-3}) \frac{d}{dt} [35 \times 10^{-3} e^{-400t}] \\ &= (100 \times 10^{-3}) (35 \times 10^{-3}) \frac{d}{dt} [e^{-400t}] \\ &= (3.5 \times 10^{-3}) (-400) e^{-400t} \end{aligned}$$

$$v_L(t) = -1.4e^{-400t} \text{ V}$$

Thus, the voltage across the inductor, v_L is $\boxed{-1.4e^{-400t} \text{ V}}$.

Step 3 of 5

(b)

As the resistor and the inductor are connected in parallel, the voltage across the resistor is the same as that of the inductor.

Thus, the voltage across the resistor, v_R is $\boxed{-1.4e^{-400t} \text{ V}}$

Step 4 of 5

(c)

Apply Kirchhoff's voltage law to the inductor-resistor loop to determine sum of the voltages across the inductor and the resistor.

$$\begin{aligned} -v_L(t) + v_R(t) &= -(-1.4e^{-400t}) - 1.4e^{-400t} \\ &= 0 \text{ V} \end{aligned}$$

Determine the sum of voltages across the resistor and the inductor.

$$\begin{aligned} v_L + v_R &= -1.4e^{-400t} - 1.4e^{-400t} \\ v_L + v_R &= -2.8e^{-400t} \end{aligned}$$

Thus, the sum of voltages across the resistor and the inductor is **not equal to 0**.

Step 5 of 5

(d)

Determine the energy in the inductor.

$$\begin{aligned} w &= \frac{1}{2} Li^2 \\ &= \frac{1}{2} (100 \times 10^{-3}) (35 \times 10^{-3} e^{-400t})^2 \\ &= \frac{1}{2} (100 \times 10^{-3}) (35 \times 10^{-3})^2 e^{-800t} \\ &= 61.25 e^{-800t} \text{ } \mu\text{J} \end{aligned}$$

Determine the energy in the inductor at $t = 0$.

$$\begin{aligned} w(0) &= 61.25 e^{(-800)(0)} \text{ } \mu\text{J} \\ &= 61.25 \text{ } \mu\text{J} \\ &= 61.25 \text{ } \mu\text{J} \end{aligned}$$

Thus, the energy in the inductor at $t = 0$ is 61.25 μJ .